Jiankun (Kevin) **Xu**

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EDUCATION

The Experimental High School Attached to Beijing Normal University Sep 2024 - Jul 2027
International Division | Beijing, China | Class of 2027

GPA: 100/100 **SAT:** 1570 (Math 800; Reading and Writing 770) **TOEFL:** 113

The High School Affiliated to Beijing Normal University Sep 2021 - Jul 2024
Junior Secondary Education | Beijing, China

SELECTED RESEARCH AND MATHEMATICAL EXPOSITION

Inverse Spectral Geometry: Hearing the Shape of a Drum Grades 11-12
Sole author | Advisors: Jian Deng and Lingyun Chang

- Studied how the Dirichlet spectrum encodes the boundary of an analytic planar domain. Constructed boundary operators and finite-matrix approximations from Riemann-map coefficients.
- Derived a regularized-determinant spectral criterion and a fixed-parameter convergence result. Tested forward spectral calculations and restricted inverse fitting on the Neumann oval.

Constraint Repair for Reaction-Diffusion PINNs Grades 10-11
Co-author with Lingyun Chang | Manuscripts under review at Computer Physics Communications and ICMLC 2027

- Developed training and post-processing methods for physics-informed neural networks (PINNs) in stiff heat-species systems, testing feasibility and accuracy across 10 random seeds.
- Separated box clipping from mean correction. In one held-out benchmark, clipping reduced upper-bound violations from 11.2% to zero, while mean correction revealed an accuracy tradeoff.

Flood Susceptibility Assessment in Fangshan, Beijing 2025-2026
Co-author with Jia Li | Researcher and conference presenter

- Combined flood and insurance records with 11 environmental factors in a convolutional neural network. Evaluated susceptibility maps through field checks and factor-sensitivity analysis.

Presented at the First Symposium on Geocomputation and System Simulation (conference: July 10-12, 2026).

Integer and Plane Partitions | Pioneer Academics May-Sep 2025
Expository course paper | Advisor: Carl Yerger, Davidson College

- Reconstructed Newman's proof of the Hardy-Ramanujan partition asymptotic, expanding the generating-function argument, contour estimates, and error control for readers with a high-school mathematics background.

ADDITIONAL ENGINEERING PROJECTS

Real World Design Challenge (RWDC) | (Vice) President Sep 2024 - May 2025

- Worked on a biplane cargo-drone concept for wildfire relief, including avionics selection, power and payload checks, and route planning. Team placed 2nd in China and 3rd globally.

ISSSP | Team Leader Jun-Nov 2025

- Developed the proposal *Ilmenite Reduction and Magnetic Separation in Space*, combining carbon reduction and magnetic separation to recover iron and titanium dioxide under microgravity. Received Global Gold.

SELECTED HONORS

AIME: 14/15 (2026). **AMC 12:** 132 (2025); AMC 10/12 Distinction.

PUMaC, B Division: Team Overall 2nd; Individual Overall 10th; Number Theory 9th; Algebra Gold Medal.

ARML Local: Team 23rd globally, National Gold; Individual Round National Gold.

John Locke Essay Competition: Commendations in History and Philosophy.

PhysicsBowl, Division 1: National Top Gold. **HiMCM:** Honorable Mention.

SDSZ: Full Academic Scholarship (Grades 10-12); Academic Contribution Award.

TECHNICAL SKILLS AND LANGUAGES

Programming and typesetting: Python, R, LaTeX. **Languages:** Chinese (native), English (fluent).